APPENDIX A
PROOFS

A. Proof of Lemma 4.4

Let Φ be a random realization consistent with ψ. Let Φ_ψ denote the realization of G_ψ in which each edge retains its live or blocked state from Φ. As shown in the proof of Theorem 4.2, Φ_ψ has the same distribution as a random realization of G_ψ, and $I_\Phi(\mathrm{dom}(\psi)\cup S) = I_\psi \cup I_{\Phi_\psi}(S)$. Thus, the newly activated nodes in U_v are $U_v^\psi \cap I_{\Phi_\psi}(S)$. By the definition of f_v^ψ and linearity of expectation,

$$\begin{aligned}\Delta(S \mid \psi) &= \textstyle\sum_{v\in V} \mathbb{E}_{\Phi\sim\psi}\big[f_v^\psi\big(U_v^\psi \cap I_{\Phi_\psi}(S)\big)\big] \\ &= \textstyle\sum_{v\in V} \mathbb{E}\big[f_v^\psi\big(U_v^\psi \cap I_{G_\psi}(S)\big)\big] \\ &= \sigma_\psi(S),\end{aligned}$$

which completes the proof. □

B. Proof of Lemma 5.3

Since $m_v^\psi(\{q\}) = \ell_v^\psi(q)$, Lemma 5.2 applies with node weights $w'(q) = \ell_q^\psi(q) + \sum_{v\in N_{out}(q)} \ell_v^\psi(q)$, yielding

$$\begin{aligned}\sigma_\psi^m(S) &= \textstyle\sum_{q\in V_\psi} w'(q)\Pr[q \in I_{G_\psi}(S)] \\ &= W'_\psi \Pr[R_\psi(q^*)\cap S \neq \emptyset].\end{aligned}$$

For the surplus part, we have

$$\begin{aligned}&\sigma_\psi^g(S) = \textstyle\sum_{v\in V} \mathbb{E}\big[g_v^\psi\big(U_v^\psi \cap I_{G_\psi}(S)\big)\big] \\ &= \textstyle\sum_{v:\,\eta_v^\psi>0} \mathbb{E}_{\Phi_\psi}\big[g_v^\psi\big(\Lambda_{\mathbb{R}_\psi(v)}(S)\big)\big] \\ &= H_\psi \textstyle\sum_{v:\,\eta_v^\psi>0} \frac{\eta_v^\psi}{H_\psi}\, \mathbb{E}_{\Phi_\psi}\Big[\frac{g_v^\psi(\Lambda_{\mathbb{R}_\psi(v)}(S))}{\eta_v^\psi}\Big] \\ &= H_\psi\, \mathbb{E}\Big[\frac{g_{v^*}^\psi(\Lambda_{\mathbb{R}_\psi(v^*)}(S))}{\eta_{v^*}^\psi}\Big],\end{aligned}$$

where the second equality uses $\Lambda_{\mathbb{R}_\psi(v)}(S) = U_v^\psi \cap I_{\Phi_\psi}(S)$, and the last equality uses $\Pr[v^* = v] = \eta_v^\psi / H_\psi$. Substituting the two expressions into $\sigma_\psi(S) = \sigma_\psi^m(S) + \sigma_\psi^g(S)$ gives Eq. (7), which completes the proof. □

C. Proof of Lemma 5.5

We determine the sample sizes of both Eq. (6) and Eq. (8) by the Chernoff bound [43], which ensures an (ϵ,δ)-estimate of the mean μ of i.i.d. samples in $[0,M]$ with $c\cdot M\ln(2/\delta)/(\epsilon^2\mu)$ samples, where $c = 2+2\epsilon/3$. Let $L = c\ln(2/\delta)/(\epsilon^2\sigma_\psi(S))$ and $M_\psi = \max_{v\in V} f_v^\psi(U_v^\psi)$. For Eq. (6), sample values lie in $[0, M_\psi]$ with mean $\sigma_\psi(S)/n$, so that it takes $nM_\psi L$ RR sequences with sources drawn uniformly from V, i.e.,

$$\mathcal{C}_6 = nM_\psi L \cdot \frac{1}{n}\sum_{v\in V} C_{seq}(v) = M_\psi L \sum_{v\in V} C_{seq}(v).$$

For Eq. (8), each hybrid sample has $X(S)$ lying in $[0,1]$ with mean $\sigma_\psi(S)/F_\psi$, so that it takes $F_\psi L$ samples. By Definition 5.4, each sample is an RR set with probability W'_ψ/F_ψ and an RR sequence with probability H_ψ/F_ψ, whose expected costs are $\bar{C}_{set}$ and $\bar{C}_{seq}$, respectively. Hence,

$$\mathcal{C}_8 = F_\psi L\Big(\frac{W'_\psi}{F_\psi}\bar{C}_{set} + \frac{H_\psi}{F_\psi}\bar{C}_{seq}\Big) = L\big(W'_\psi\bar{C}_{set} + H_\psi\bar{C}_{seq}\big).$$

Since $M_\psi \geq f_v^\psi(U_v^\psi)$ for each $v\in V$, we have

$$\frac{\mathcal{C}_6}{\mathcal{C}_8} = \frac{\sum_{v\in V} M_\psi C_{seq}(v)}{W'_\psi\bar{C}_{set} + H_\psi\bar{C}_{seq}} \geq \frac{\sum_{v\in V} f_v^\psi(U_v^\psi)C_{seq}(v)}{W'_\psi\bar{C}_{set} + H_\psi\bar{C}_{seq}}.$$

Intuitively, since $f_v^\psi(U_v^\psi) = m_v^\psi(U_v^\psi)+\eta_v^\psi$, $\sum_{v\in V} m_v^\psi(U_v^\psi) = W'_\psi$, and $\bar{C}_{seq} = \sum_{v\in V}\eta_v^\psi C_{seq}(v)/H_\psi$ by Lemma 5.3, the numerator satisfies

$$\begin{aligned}\sum_{v\in V} f_v^\psi(U_v^\psi)C_{seq}(v) &= W'_\psi \sum_{v\in V}\frac{m_v^\psi(U_v^\psi)}{W'_\psi}C_{seq}(v) + H_\psi\bar{C}_{seq} \\ &\approx W'_\psi\sum_{v\in V}\frac{\eta_v^\psi}{H_\psi}C_{seq}(v) + H_\psi\bar{C}_{seq} \\ &= W'_\psi\bar{C}_{seq} + H_\psi\bar{C}_{seq} = F_\psi\bar{C}_{seq}.\end{aligned}$$

As $W'_\psi/F_\psi = 1-\beta_\psi$, the bound then becomes

$$\begin{aligned}&\frac{\sum_{v\in V} f_v^\psi(U_v^\psi)C_{seq}(v)}{W'_\psi\bar{C}_{set} + H_\psi\bar{C}_{seq}} \approx \frac{F_\psi\bar{C}_{seq}}{W'_\psi\bar{C}_{set} + H_\psi\bar{C}_{seq}} \\ &= \frac{1}{\beta_\psi + (1-\beta_\psi)\bar{C}_{set}/\bar{C}_{seq}} \approx \min\Big\{\frac{1}{\beta_\psi}, \frac{\bar{C}_{seq}}{\bar{C}_{set}}\Big\},\end{aligned}$$

where the last step holds as the denominator lies between $\max\{\beta_\psi, \bar{C}_{set}/\bar{C}_{seq}\}$ and twice of it. □

D. Proof of Lemma 5.6

Consider a hybrid sample $\mathbb{H}$ with value $X(\cdot)$. If $\mathbb{H}$ is an RR set $R_\psi(q^*)$, then $X(S) = \mathbf{1}_{\{R_\psi(q^*)\cap S\neq\emptyset\}}$ is a coverage function, and thus monotone and submodular. Otherwise, $\mathbb{H}$ is an RR sequence $\mathbb{R}_\psi(v)$ with $X(S) = g_v^\psi(\Lambda_{\mathbb{R}_\psi(v)}(S))/\eta_v^\psi$. For any $x\in U_v^\psi$ and $B\subseteq B'\subseteq U_v^\psi\setminus\{x\}$, we have $A_v^\psi\cup B \subseteq A_v^\psi\cup B'$, so that the submodularity of f_v gives

$$\begin{aligned}f_v^\psi(B\cup\{x\}) - f_v^\psi(B) &= f_v(A_v^\psi\cup B\cup\{x\}) - f_v(A_v^\psi\cup B) \\ &\geq f_v(A_v^\psi\cup B'\cup\{x\}) - f_v(A_v^\psi\cup B') \\ &= f_v^\psi(B'\cup\{x\}) - f_v^\psi(B') \\ &\geq f_v^\psi(U_v^\psi) - f_v^\psi(U_v^\psi\setminus\{x\}) = \ell_v^\psi(x),\end{aligned}$$

i.e., f_v^ψ is submodular, and the marginal gain of x is at least $\ell_v^\psi(x)$. As $m_v^\psi(B\cup\{x\}) - m_v^\psi(B) = \ell_v^\psi(x)$, we have

$$\begin{aligned}&g_v^\psi(B\cup\{x\}) - g_v^\psi(B) \\ &= f_v^\psi(B\cup\{x\}) - f_v^\psi(B) - \big(m_v^\psi(B\cup\{x\}) - m_v^\psi(B)\big) \\ &= f_v^\psi(B\cup\{x\}) - f_v^\psi(B) - \ell_v^\psi(x) \geq 0,\end{aligned}$$

i.e., g_v^ψ is monotone. Moreover, $g_v^\psi = f_v^\psi - m_v^\psi$ is submodular, as f_v^ψ is submodular and m_v^ψ is modular. Finally, for each $s\in V_\psi$, let $T_s = \{u\in U_v^\psi : s\in R_\psi(u)\}$, so that $\Lambda_{\mathbb{R}_\psi(v)}(S) = \bigcup_{s\in S} T_s$ grows with S. Hence, for any $S\subseteq S'$ and $s\notin S'$,

$$\begin{aligned}&g_v^\psi\big(\Lambda_{\mathbb{R}_\psi(v)}(S)\cup T_s\big) - g_v^\psi\big(\Lambda_{\mathbb{R}_\psi(v)}(S)\big) \\ &\quad\geq g_v^\psi\big(\Lambda_{\mathbb{R}_\psi(v)}(S')\cup T_s\big) - g_v^\psi\big(\Lambda_{\mathbb{R}_\psi(v)}(S')\big) \geq 0,\end{aligned}$$

where the two inequalities follow from the submodularity and monotonicity of g_v^ψ, respectively. Hence, $X(\cdot)$ is monotone and submodular in both cases, and so is $\Gamma_{\mathcal{H}}(\cdot)$ as a sum of such functions, which completes the proof. □

E. Concentration Bounds for Hybrid Samples

The sampling schedule of MODE relies on the following concentration bounds for hybrid samples, adapted from those used in IM solutions [10].

Lemma A.1. *Given a seed set S and a collection $\mathcal{H}$ of θ independent random hybrid samples on G_ψ, let $\mu = \theta \cdot \sigma_\psi(S)/F_\psi$. For any $\lambda > 0$, we have*

$$\Pr\left[\Gamma_{\mathcal{H}}(S) - \mu \geq \lambda\right] \leq \exp\left(-\frac{\lambda^2}{2\lambda/3 + 2\mu}\right), \tag{14}$$

$$\Pr\left[\Gamma_{\mathcal{H}}(S) - \mu \leq -\lambda\right] \leq \exp\left(-\frac{\lambda^2}{2\mu}\right). \tag{15}$$

Proof. A sequence of random variables $M_1, M_2, \ldots, M_k$ is a *martingale* if and only if $\mathbb{E}[|M_j|] < +\infty$ and $\mathbb{E}[M_j \mid M_1, \ldots, M_{j-1}] = M_{j-1}$ for all $2 \leq j \leq k$. Our proof relies on the following well-established concentration result for martingales [43].

Lemma A.2. *Let $M_1, M_2, \ldots, M_k$ be a martingale such that $M_1 - \mathbb{E}[M_1] \leq a$ and $M_j - M_{j-1} \leq a$ for any $j \in [2, k]$, and let b satisfy $\mathit{Var}[M_1] + \sum_{j=2}^{k} \mathit{Var}[M_j \mid M_1, \ldots, M_{j-1}] \leq b$. Then, for any $\lambda > 0$,*

$$\Pr[M_k - \mathbb{E}[M_k] \geq \lambda] \leq \exp\left(-\frac{\lambda^2}{2a\lambda/3 + 2b}\right), \tag{16}$$

where $\mathit{Var}[\cdot]$ denotes the variance of a random variable.

For the j-th hybrid sample $\mathbb{H}_j \in \mathcal{H}$, let $Y_j = X_j(S)$ and $p = \sigma_\psi(S)/F_\psi$, and define $Z_j = \sum_{t=1}^{j}(Y_t - p)$, so that $Z_\theta = \Gamma_{\mathcal{H}}(S) - \mu$. By Eq. (8), $\mathbb{E}[Y_j] = p$, and since the samples are generated independently, $\mathbb{E}[Y_j \mid Y_1, \ldots, Y_{j-1}] = \mathbb{E}[Y_j] = p$. Hence, $\mathbb{E}[Z_j \mid Z_1, \ldots, Z_{j-1}] = Z_{j-1}$, i.e., $\{Z_j\}_{j=1}^{\theta}$ forms a martingale. Since $Y_j \in [0, 1]$, its conditional variances satisfy

$$\begin{aligned}
&\mathrm{Var}[Z_1] + \textstyle\sum_{j=2}^{\theta} \mathrm{Var}[Z_j \mid Z_1, \ldots, Z_{j-1}] \\
&= \textstyle\sum_{j=1}^{\theta} \mathrm{Var}[Y_j] \leq \sum_{j=1}^{\theta} \left(\mathbb{E}[Y_j] - \mathbb{E}[Y_j]^2\right) \\
&= \theta p(1-p) \leq \theta p.
\end{aligned}$$

For the upper tail, we have $Z_1 - \mathbb{E}[Z_1] = Z_1 \leq 1$ and $Z_j - Z_{j-1} \leq 1$, and setting $M_j = Z_j$, $a = 1$, and $b = \theta p = \mu$ in Eq. (16) gives Eq. (14). For the lower tail, $\{-Z_j\}_{j=1}^{\theta}$ is also a martingale with $-Z_1 \leq p$ and $-Z_j + Z_{j-1} \leq p$, and since negation does not change variances, its conditional variances sum to at most $\theta p(1-p)$ by the above inequality. Setting $M_j = -Z_j$, $a = p$, and $b = \theta p(1-p)$ in Eq. (16) gives

$$\Pr[\Gamma_{\mathcal{H}}(S) - \mu \leq -\lambda] \leq \exp\Big(-\frac{\lambda^2}{2p\lambda/3 + 2\theta p(1-p)}\Big).$$

If $\lambda > \mu$, the event $\Gamma_{\mathcal{H}}(S) \leq \mu - \lambda < 0$ is impossible since $\Gamma_{\mathcal{H}}(S) \geq 0$, and Eq. (15) holds trivially. Otherwise, $\lambda \leq \mu = \theta p$ implies $2p\lambda/3 + 2\theta p(1-p) \leq 2\theta p^2/3 + 2\theta p - 2\theta p^2 \leq 2\mu$, and Eq. (15) follows. □

F. Proof of Theorem 5.7

The proof adapts the analysis of OPIM-C [9], where the concentration bounds for RR sets are replaced by those for hybrid samples (Lemma A.1). Since θ starts from $\theta_0 = \lceil \theta_{\max} \cdot \mathsf{LB}_\psi / F_\psi \rceil$ and doubles in each iteration until it reaches $\theta_{\max}$, MODE runs at most $\lceil \log_2(\theta_{\max}/\theta_0) \rceil + 1 \leq T$ iterations, where the sample size θ_t of the t-th iteration is fixed in advance, and $\mathcal{H}_1$ and $\mathcal{H}_2$ consist of the first θ_t samples of two independent sample streams. Following the proof sketch, we first show that σ^u and σ^l are valid bounds in every iteration and that the estimation errors are small at $\theta_{\max}$, all with high probability, and then derive the guarantee on these events.

Upper bound σ^u. For each iteration t, let $\mathcal{E}_1^t$ be the event that $\sigma^u \geq \mathrm{OPT}_\psi$, and $\mu^o = \theta_t \mathrm{OPT}_\psi / F_\psi$. Since S_ψ^o is a fixed set, setting $S = S_\psi^o$ and $\lambda = \sqrt{2a\mu^o}$ in Eq. (15) gives

$$\Pr\left[\Gamma_{\mathcal{H}_1}(S_\psi^o) \leq \mu^o - \sqrt{2a\mu^o}\right] \leq \exp\Big(-\frac{2a\mu^o}{2\mu^o}\Big) = e^{-a}.$$

Meanwhile, by the monotonicity and submodularity of $\Gamma_{\mathcal{H}_1}(\cdot)$ (Lemma 5.6) and $|S_\psi^o| = b$, we have

$$\begin{aligned}
\Gamma_{\mathcal{H}_1}(S_\psi^o) &\leq \min_{0 \leq i \leq b} \Big(\Gamma_{\mathcal{H}_1}(S_i) + \sum_{x \in S_\psi^o} \Gamma_{\mathcal{H}_1}(x \mid S_i)\Big) \\
&\leq \min_{0 \leq i \leq b} \Big(\Gamma_{\mathcal{H}_1}(S_i) + \max_{Q \subseteq V_\psi, |Q| \leq b} \sum_{x \in Q} \Gamma_{\mathcal{H}_1}(x \mid S_i)\Big) \\
&= \Gamma^u
\end{aligned}$$

by Eq. (13). Thus, $\mu^o - \sqrt{2a\mu^o} < \Gamma^u$ with probability at least $1 - e^{-a}$, and solving this quadratic inequality in $\sqrt{\mu^o}$ gives

$$\mathrm{OPT}_\psi = \frac{F_\psi \mu^o}{\theta_t} < \Big(\sqrt{\Gamma^u + \frac{a}{2}} + \sqrt{\frac{a}{2}}\Big)^2 \cdot \frac{F_\psi}{\theta_t} = \sigma^u$$

by Eq. (11). Hence, $\Pr[\mathcal{E}_1^t] \geq 1 - e^{-a}$.

Lower bound σ^l. For each iteration t, let $\mathcal{E}_2^t$ be the event that $\sigma^l \leq \sigma_\psi(S)$ for the seed set S selected in this iteration, and $\mu = \theta_t \sigma_\psi(S)/F_\psi$. Since S is determined by $\mathcal{H}_1$ and thus independent of $\mathcal{H}_2$, setting $\lambda = a/3 + y$ with $y = \sqrt{a^2/9 + 2a\mu}$ in Eq. (14), which satisfies $\lambda^2 = (2\lambda/3 + 2\mu)a$, gives

$$\Pr\Big[\Gamma_{\mathcal{H}_2}(S) \geq \mu + \frac{a}{3} + y\Big] \leq \exp\Big(-\frac{\lambda^2}{2\lambda/3 + 2\mu}\Big) = e^{-a}.$$

Meanwhile, as $\mu = (y^2 - a^2/9)/(2a)$ and $y \geq 0$, we have

$$\begin{aligned}
\Gamma_{\mathcal{H}_2}(S) < \mu + \frac{a}{3} + y &\iff (y+a)^2 > 2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9} \\
&\iff y > \sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}} - a.
\end{aligned}$$

If $\Gamma_{\mathcal{H}_2}(S) > 2a/3$, then

$$y > \sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}} - a > \sqrt{\frac{4a^2}{3} + \frac{4a^2}{9}} - a = \frac{a}{3} > 0.$$

Hence, $y^2 > \left(\sqrt{2a\Gamma_{\mathcal{H}_2}(S) + 4a^2/9} - a\right)^2$, and thus

$$\begin{aligned}\mu = \frac{y^2 - a^2/9}{2a} &> \frac{1}{2a}\left(\left(\sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}} - a\right)^2 - \frac{a^2}{9}\right)\\ &= \Gamma_{\mathcal{H}_2}(S) + \frac{2a}{3} - \sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}}\\ &= \left(\sqrt{\Gamma_{\mathcal{H}_2}(S) + \frac{2a}{9}} - \sqrt{\frac{a}{2}}\right)^2 - \frac{a}{18} = \frac{\theta_t\sigma^l}{F_\psi}\end{aligned}$$

by Eq. (12), i.e., $\sigma_\psi(S) > \sigma^l$. Otherwise, $\Gamma_{\mathcal{H}_2}(S) \le 2a/3$, and since $\Gamma_{\mathcal{H}_2}(S) \ge 0$, we have

$$\begin{aligned}&\left(\Gamma_{\mathcal{H}_2}(S) + \frac{2a}{3}\right)^2 - \left(2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}\right)\\ &= \Gamma_{\mathcal{H}_2}(S)\left(\Gamma_{\mathcal{H}_2}(S) - \frac{2a}{3}\right) \le 0,\end{aligned}$$

which, by the above equalities, gives

$$\frac{\theta_t\sigma^l}{F_\psi} = \Gamma_{\mathcal{H}_2}(S) + \frac{2a}{3} - \sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}} \le 0,$$

i.e., $\sigma^l \le 0 \le \sigma_\psi(S)$. Hence, $\Pr[\mathcal{E}_2^t] \ge \Pr[\Gamma_{\mathcal{H}_2}(S) < \mu + a/3 + y] \ge 1 - e^{-a}$.

Estimation errors at $\theta_{\max}$. Let $\mathcal{E}_3$ be the event that, when $|\mathcal{H}_1| = \theta_{\max}$, every size-$b$ set $S' \subseteq V_\psi$ satisfies $|F_\psi\Gamma_{\mathcal{H}_1}(S')/\theta_{\max} - \sigma_\psi(S')| \le \epsilon_c\mathrm{OPT}_\psi$, i.e., $|\Gamma_{\mathcal{H}_1}(S') - \mu_{S'}| \le \epsilon_c\mu^*$, where $\mu_{S'} = \theta_{\max}\sigma_\psi(S')/F_\psi$ and $\mu^* = \theta_{\max}\mathrm{OPT}_\psi/F_\psi \ge \mu_{S'}$. Fix such a set S'. Setting $S = S'$ and $\lambda = \epsilon_c\mu^*$ in Eqs. (14)-(15) gives

$$\begin{aligned}&\Pr\left[|\Gamma_{\mathcal{H}_1}(S') - \mu_{S'}| > \epsilon_c\mu^*\right]\\ &\le \exp\Big(-\frac{\epsilon_c^2\mu^{*2}}{2\epsilon_c\mu^*/3 + 2\mu_{S'}}\Big) + \exp\Big(-\frac{\epsilon_c^2\mu^{*2}}{2\mu_{S'}}\Big)\\ &\le 2\exp\Big(-\frac{\epsilon_c^2\mu^*}{2 + 2\epsilon_c/3}\Big) = 2\exp\Big(-\frac{\epsilon_c^2\theta_{\max}\mathrm{OPT}_\psi}{(2+2\epsilon_c/3)F_\psi}\Big)\\ &\le 2\exp\Big(-\frac{\mathrm{OPT}_\psi}{\mathsf{LB}_\psi}\ln\frac{8\binom{|V_\psi|}{b}}{\delta}\Big) \le \frac{\delta}{4\binom{|V_\psi|}{b}},\end{aligned}$$

where the second inequality uses $\mu_{S'} \le \mu^*$, the third follows from Eq. (9), and the last follows from $\mathsf{LB}_\psi \le \mathrm{OPT}_\psi$, as seeding any single node q yields an adoption value of at least $f_v^\psi(\{q\})$ for each v with $q \in U_v^\psi$. By the union bound over all $\binom{|V_\psi|}{b}$ size-b sets, $\Pr[\mathcal{E}_3] \ge 1 - \delta/4$.

Approximation guarantee. Since $a = \ln(4T/\delta)$, the union bound over at most T iterations shows that all events $\{\mathcal{E}_1^t, \mathcal{E}_2^t\}_t$ and $\mathcal{E}_3$ occur simultaneously with probability at least $1 - 2T\cdot\delta/(4T) - \delta/4 \ge 1 - \delta$. On this event, consider the iteration in which MODE terminates. If the ratio test succeeds, then

$$\sigma_\psi(S) \ge \sigma^l \ge \rho_b(1-\epsilon')\sigma^u \ge \rho_b(1-\epsilon')\mathrm{OPT}_\psi.$$

Otherwise, $|\mathcal{H}_1| = \theta_{\max}$, and the greedy algorithm achieves a ρ_b-approximation for maximizing the monotone submodular function $\Gamma_{\mathcal{H}_1}(\cdot)$ [42], so that

$$\begin{aligned}\sigma_\psi(S) &\ge F_\psi\Gamma_{\mathcal{H}_1}(S)/\theta_{\max} - \epsilon_c\mathrm{OPT}_\psi\\ &\ge \rho_b F_\psi\Gamma_{\mathcal{H}_1}(S^o_\psi)/\theta_{\max} - \epsilon_c\mathrm{OPT}_\psi\\ &\ge \rho_b(1-\epsilon_c)\mathrm{OPT}_\psi - \epsilon_c\mathrm{OPT}_\psi = \rho_b(1-\epsilon')\mathrm{OPT}_\psi,\end{aligned}$$

where the first and last inequalities follow from $\mathcal{E}_3$, and the equality follows from $\epsilon_c = \rho_b\epsilon'/(1+\rho_b)$. Hence, $\Pr[\sigma_\psi(S) \ge \rho_b(1-\epsilon')\mathrm{OPT}_\psi] \ge 1 - \delta$. Finally, since $\sigma_\psi(S) \ge 0$ always holds, taking the expectation over the randomness ω gives

$$\mathbb{E}_\omega[\sigma_\psi(S)] \ge (1-\delta)(1-\epsilon')\rho_b\mathrm{OPT}_\psi = \rho_b(1-\epsilon)\mathrm{OPT}_\psi,$$

where the equality holds by the choice $\epsilon' = (\epsilon - \delta)/(1-\delta)$, which completes the proof. □

G. Proof of Theorem 5.8

The running time of MODE consists of the costs of sample generation, seed selection, and bound computation. As all three costs grow with the number of samples, we first bound the expected number of samples, and then each cost.

Number of samples. Let

$$\theta^* = \frac{c\cdot F_\psi\big(\ln\binom{|V_\psi|}{b} + \ln(T/\delta)\big)}{\epsilon'^2\mathrm{OPT}_\psi}$$

for a sufficiently large constant c. Following the analysis of OPIM-C [9], we first show that in any iteration t with $\theta_t \ge \theta^*$,

$$\Pr\left[\sigma^l/\sigma^u \ge \rho_b(1-\epsilon')\right] \ge 1 - \delta/T.$$

For each iteration t, let $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ be the events that

$$\begin{aligned}&\mathcal{A}_1 : \max_{S'\subseteq V_\psi, |S'|=b}\big(F_\psi\Gamma_{\mathcal{H}_1}(S')/\theta_t - \sigma_\psi(S')\big) \le \epsilon'\mathrm{OPT}_\psi/8,\\ &\mathcal{A}_2 : F_\psi\Gamma_{\mathcal{H}_1}(S^o_\psi)/\theta_t \ge (1-\epsilon'/8)\mathrm{OPT}_\psi,\\ &\mathcal{A}_3 : F_\psi\Gamma_{\mathcal{H}_2}(S)/\theta_t \ge \sigma_\psi(S) - \epsilon'\mathrm{OPT}_\psi/8,\end{aligned}$$

where S is the seed set selected in the t-th iteration. As the means of $\Gamma_{\mathcal{H}_1}(S')$, $\Gamma_{\mathcal{H}_1}(S^o_\psi)$, and $\Gamma_{\mathcal{H}_2}(S)$ are at most $\theta_t\mathrm{OPT}_\psi/F_\psi$, setting $\lambda = \epsilon'\theta_t\mathrm{OPT}_\psi/(8F_\psi)$ in Lemma A.1 shows that each of $\mathcal{A}_2$, $\mathcal{A}_3$, and the inequality in $\mathcal{A}_1$ for a fixed S' fails with probability at most

$$\exp\Big(-\frac{\lambda^2}{2\lambda/3 + 2\theta_t\mathrm{OPT}_\psi/F_\psi}\Big), \tag{17}$$

which is at most $\delta/(3T\binom{|V_\psi|}{b})$ if $\theta_t \ge \theta^*$ for a sufficiently large c. Taking the union bound over all $\binom{|V_\psi|}{b}$ size-b sets S' in $\mathcal{A}_1$ thus gives

$$\Pr[\mathcal{A}_1\cap\mathcal{A}_2\cap\mathcal{A}_3] \ge 1 - \left(\binom{|V_\psi|}{b} + 2\right)\frac{\delta}{3T\binom{|V_\psi|}{b}} \ge 1 - \frac{\delta}{T}.$$

Meanwhile, for each $0 \le i < b$, the greedy choice gives

$$\begin{aligned}\Gamma_{\mathcal{H}_1}(S_{i+1}) - \Gamma_{\mathcal{H}_1}(S_i) &= \max_{x\in V_\psi}\Gamma_{\mathcal{H}_1}(x \mid S_i)\\ &\ge \frac{1}{b}\max_{Q\subseteq V_\psi, |Q|\le b}\sum_{x\in Q}\Gamma_{\mathcal{H}_1}(x\mid S_i)\\ &\ge \frac{1}{b}\big(\Gamma^u - \Gamma_{\mathcal{H}_1}(S_i)\big),\end{aligned}$$

where the last inequality follows from Eq. (13). Rearranging it as $\Gamma^u - \Gamma_{\mathcal{H}_1}(S_{i+1}) \le (1-1/b)(\Gamma^u - \Gamma_{\mathcal{H}_1}(S_i))$ and applying it for $i = b-1, \ldots, 0$ with $\Gamma_{\mathcal{H}_1}(\emptyset) = 0$ yield

$$\begin{aligned}\Gamma^u - \Gamma_{\mathcal{H}_1}(S) &\le \Big(1 - \frac{1}{b}\Big)\big(\Gamma^u - \Gamma_{\mathcal{H}_1}(S_{b-1})\big)\\ &\le \cdots \le \Big(1 - \frac{1}{b}\Big)^b\Gamma^u,\end{aligned}$$

which, as $\rho_b = 1 - (1 - 1/b)^b$, gives

$$\Gamma_{\mathcal{H}_1}(S) \geq \rho_b \Gamma^u. \tag{18}$$

Moreover, on these events, $\mathcal{A}_2$ and $\epsilon' < 1$ give

$$\Gamma^u \geq \Gamma_{\mathcal{H}_1}(S^o_\psi) \geq 7\theta_t \mathrm{OPT}_\psi/(8F_\psi), \tag{19}$$

which further gives

$$\begin{aligned}
\Gamma_{\mathcal{H}_2}(S) &\geq \frac{\theta_t}{F_\psi}\Big(\sigma_\psi(S) - \frac{\epsilon' \mathrm{OPT}_\psi}{8}\Big) \\
&\geq \frac{\theta_t}{F_\psi}\Big(\frac{F_\psi \Gamma_{\mathcal{H}_1}(S)}{\theta_t} - \frac{\epsilon' \mathrm{OPT}_\psi}{4}\Big) \\
&\geq \frac{\theta_t}{F_\psi}\Big(\frac{\rho_b F_\psi \Gamma^u}{\theta_t} - \frac{\epsilon' \mathrm{OPT}_\psi}{4}\Big) \\
&\geq \Big(\frac{7\rho_b}{8} - \frac{1}{4}\Big)\frac{\theta_t \mathrm{OPT}_\psi}{F_\psi} \geq \frac{\theta_t \mathrm{OPT}_\psi}{4F_\psi},
\end{aligned}$$

where the inequalities follow, in turn, from $\mathcal{A}_3$, $\mathcal{A}_1$, Eq. (18), Eq. (19) and $\epsilon' < 1$, and $\rho_b \geq 1 - 1/e$. By Eq. (19), both Γ^u and $\Gamma_{\mathcal{H}_2}(S)$ are thus at least

$$\frac{\theta_t \mathrm{OPT}_\psi}{4F_\psi} \geq \frac{\theta^* \mathrm{OPT}_\psi}{4F_\psi} \geq \frac{c \ln(T/\delta)}{4\epsilon'^2} \geq \frac{c\, a}{12\epsilon'^2},$$

where the inequalities follow, in turn, from $\theta_t \geq \theta^*$, the definition of θ^*, and $a = \ln(4T/\delta) \leq 3\ln(T/\delta)$ as $T/\delta \geq 2$. Hence, both a/Γ^u and $a/\Gamma_{\mathcal{H}_2}(S)$ are at most $12\epsilon'^2/c \leq (\epsilon'/24)^2$ for a sufficiently large c, and expanding Eqs. (11)-(12) gives

$$\begin{aligned}
\frac{\theta_t \sigma^u}{F_\psi} &= \Gamma^u + a + \sqrt{2a\Gamma^u + a^2} \leq \Gamma^u + 3\sqrt{a\Gamma^u} \\
&\leq \Big(1 + \frac{\epsilon'}{8}\Big)\Gamma^u, \\
\frac{\theta_t \sigma^l}{F_\psi} &= \Gamma_{\mathcal{H}_2}(S) + \frac{2a}{3} - \sqrt{2a\Gamma_{\mathcal{H}_2}(S) + \frac{4a^2}{9}} \\
&\geq \Gamma_{\mathcal{H}_2}(S) - \sqrt{3a\Gamma_{\mathcal{H}_2}(S)} \geq \Big(1 - \frac{\epsilon'}{8}\Big)\Gamma_{\mathcal{H}_2}(S).
\end{aligned}$$

Combining these bounds gives

$$\begin{aligned}
\frac{\sigma^l}{\sigma^u} &\geq \Big(1 - \frac{\epsilon'}{4}\Big) \cdot \frac{\Gamma_{\mathcal{H}_2}(S)}{\Gamma^u} \\
&\geq \Big(1 - \frac{\epsilon'}{4}\Big) \cdot \frac{F_\psi \Gamma_{\mathcal{H}_1}(S)/\theta_t - \epsilon' \mathrm{OPT}_\psi/4}{F_\psi \Gamma^u/\theta_t} \\
&\geq \Big(1 - \frac{\epsilon'}{4}\Big)\Big(\rho_b - \frac{2\epsilon'}{7}\Big) \geq \rho_b - \Big(\frac{1}{4} + \frac{2}{7}\Big)\epsilon' \\
&\geq \rho_b(1 - \epsilon'),
\end{aligned}$$

where the second inequality follows from $\mathcal{A}_3$ and $\mathcal{A}_1$, the third follows from Eqs. (18)-(19), the fourth uses $\rho_b \leq 1$, and the last uses $1/4 + 2/7 < 1 - 1/e \leq \rho_b$. Hence, $\Pr[\sigma^l/\sigma^u \geq \rho_b(1 - \epsilon')] \geq 1 - \delta/T$ in any iteration t with $\theta_t \geq \theta^*$.

Moreover, if $\theta_t \geq 2^j\theta^*$, we have

$$\begin{aligned}
&\Pr\left[\sigma^l/\sigma^u < \rho_b(1 - \epsilon')\right] \leq \Pr[\bar{\mathcal{A}}_1] + \Pr[\bar{\mathcal{A}}_2] + \Pr[\bar{\mathcal{A}}_3] \\
&\leq \Big(\binom{|V_\psi|}{b} + 2\Big)\exp\Big(-\frac{(\epsilon'/8)^2}{2 + \epsilon'/12} \cdot \frac{\theta_t \mathrm{OPT}_\psi}{F_\psi}\Big) \\
&\leq \Big(\binom{|V_\psi|}{b} + 2\Big)\Big(\exp\Big(-\frac{(\epsilon'/8)^2}{2 + \epsilon'/12} \cdot \frac{\theta^* \mathrm{OPT}_\psi}{F_\psi}\Big)\Big)^{2^j} \\
&\leq \Big(\binom{|V_\psi|}{b} + 2\Big)\Big(\frac{\delta}{3T\binom{|V_\psi|}{b}}\Big)^{2^j} \leq \Big(\frac{\delta}{T}\Big)^{2^j},
\end{aligned}$$

where $\bar{\mathcal{A}}_i$ is the complement of $\mathcal{A}_i$, the first inequality holds as the ratio test succeeds on $\mathcal{A}_1 \cap \mathcal{A}_2 \cap \mathcal{A}_3$, the second follows from Eq. (17) with $\lambda = \epsilon'\theta_t \mathrm{OPT}_\psi/(8F_\psi)$ and the union bound over the sets in $\mathcal{A}_1$, the third uses $\theta_t \geq 2^j\theta^*$, and the fourth holds as Eq. (17) at $\theta_t = \theta^*$ is at most $\delta/(3T\binom{|V_\psi|}{b})$.

Now, let t^* be the first iteration at which θ_t reaches θ^*. For each $j \geq 1$, `MODE` enters the $(t^* + j)$-th iteration only if the ratio test fails in the previous one, which happens with probability at most $(\delta/T)^{2^{j-1}}$ by the above bound. If `MODE` stops before the t^*-th iteration, fewer than $2\theta^*$ samples are drawn. Otherwise, since $\mathcal{H}_1$ and $\mathcal{H}_2$ consist of the first θ_t samples of two streams, the total number N of samples is twice the final sample size, so that

$$\begin{aligned}
\mathbb{E}[N] &\leq 2\theta_{t^*} + \sum_{j \geq 1} 2\theta_{t^*+j}\Big(\frac{\delta}{T}\Big)^{2^{j-1}} \\
&\leq 2\theta_{t^*}\Big(1 + \sum_{j \geq 1} 2^j\Big(\frac{\delta}{T}\Big)^{2^{j-1}}\Big) \\
&\leq 2\theta_{t^*}\Big(1 + \sum_{j \geq 1} 2^{j - 2^{j-1}}\Big) \leq 8\theta_{t^*} < 16\theta^*,
\end{aligned}$$

where the second inequality uses $\theta_{t^*+j} \leq 2^j\theta_{t^*}$, the third uses $\delta/T \leq 1/2$, the fourth uses $\sum_{j \geq 1} 2^{j - 2^{j-1}} \leq 1 + 1 + \sum_{j \geq 3} 2^{2-j} = 3$, and the last uses $\theta_{t^*} < 2\theta^*$, as θ_t doubles in each iteration from $\theta_0 \leq \theta^*$ for a sufficiently large c. Therefore, $\mathbb{E}[N] < 16\theta^* = O(\Upsilon_\psi F_\psi/\mathrm{OPT}_\psi)$ in both cases.

Sample generation. As each sample is an RR set with probability W'_ψ/F_ψ and an RR sequence with probability H_ψ/F_ψ, with the expected costs $\bar{C}_{set}$ and $\bar{C}_{seq}$ in Lemma 5.5, Wald's equation gives an expected generation cost of

$$\begin{aligned}
&\mathbb{E}[N]\Big(\frac{W'_\psi}{F_\psi}\bar{C}_{set} + \frac{H_\psi}{F_\psi}\bar{C}_{seq}\Big) \\
&= O\Big(\frac{\Upsilon_\psi(W'_\psi \bar{C}_{set} + H_\psi \bar{C}_{seq})}{\mathrm{OPT}_\psi}\Big).
\end{aligned}$$

For RR sets, a node y is visited, with its in-edges examined, if and only if the source is reachable from y, so that

$$\begin{aligned}
W'_\psi \bar{C}_{set} &= \textstyle\sum_{q \in V_\psi} w'(q) \sum_{y \in V_\psi}(1 + d_y) \Pr[q \in I_{G_\psi}(\{y\})] \\
&= \textstyle\sum_{y \in V_\psi}(1 + d_y)\, \sigma^m_\psi(\{y\}) \leq (|V_\psi| + |E_\psi|)\mathrm{OPT}_\psi,
\end{aligned}$$

where d_y is the in-degree of y in G_ψ, the second equality follows from Lemma 5.2 applied to $\{m^\psi_v\}_{v \in V}$, and the inequality follows from $\sigma^m_\psi(\{y\}) \leq \sigma_\psi(\{y\}) \leq \mathrm{OPT}_\psi$. For RR sequences, generating $\mathbb{R}_\psi(v^*)$ traverses the RR sets of all

nodes in $U^{\psi}_{v^*}$ simultaneously, where each source in $\Lambda(z)$ needs to be propagated over the in-edges of z only once, so that its cost is at most that of generating these RR sets separately, i.e.,

$$\begin{aligned}\bar{C}_{seq} &= O\Big(\max_{v^* \in V} |U^{\psi}_{v^*}| \cdot (|V_\psi| + |E_\psi|)\Big) \\ &= O\big(d^{\max}_{in}(|V_\psi| + |E_\psi|)\big),\end{aligned}$$

where the equality uses $|U^{\psi}_{v^*}| \leq |U_{v^*}| \leq 1 + d^{\max}_{in} \leq 2d^{\max}_{in}$. Plugging these two bounds into the above generation cost gives

$$O\Big(\Upsilon_\psi(|V_\psi| + |E_\psi|) + \frac{\Upsilon_\psi d^{\max}_{in}(|V_\psi| + |E_\psi|)H_\psi}{\mathrm{OPT}_\psi}\Big).$$

Seed selection and bound computation. Both steps scan the samples in each iteration: Algorithm 4 processes each RR set once and each RR sequence at most b times, each in time linear in its size, which never exceeds its generation cost, and computing σ^u additionally takes $O(b|V_\psi|)$ time to find the b largest marginal gains w.r.t. each S_i. Hence, the expected cost of these two steps is

$$\begin{aligned}&O\Big(\theta^* \cdot \frac{W'_\psi \bar{C}_{set} + bH_\psi \bar{C}_{seq}}{F_\psi} + Tb|V_\psi|\Big) \\ &= O\Big(\Upsilon_\psi\Big(|V_\psi| + |E_\psi| + \frac{bH_\psi \bar{C}_{seq}}{\mathrm{OPT}_\psi}\Big) + b|V_\psi| \log \frac{F_\psi}{\mathsf{LB}_\psi}\Big)\end{aligned}$$

where the first bound follows as for $\mathbb{E}[N]$. Summing it up with the generation cost completes the proof. □

H. Proof of Theorem 6.1

Since $\bar{\sigma}$ is induced by pointwise upper bounds, we have $\bar{\sigma}(\pi) \geq \sigma_{\text{avg}}(\pi)$ for any policy π, and thus $\bar{\sigma}(\pi^o) \geq \sigma_{\text{avg}}(\pi^o)$. Moreover, since the surrogate functions are monotone submodular, optimizing $\bar{\sigma}$ via `AdaMODE` yields a policy π with an expected $(1 - e^{\rho_b(\epsilon-1)})$-approximation w.r.t. $\bar{\sigma}(\bar{\pi}^o)$, where $\bar{\pi}^o$ is the optimal policy in Π_b for maximizing $\bar{\sigma}$. Thus,

$$\begin{aligned}\mathbb{E}_\omega[\bar{\sigma}(\pi(\omega))] &\geq (1 - e^{\rho_b(\epsilon-1)})\bar{\sigma}(\bar{\pi}^o) \\ &\geq (1 - e^{\rho_b(\epsilon-1)})\bar{\sigma}(\pi^o) \\ &\geq (1 - e^{\rho_b(\epsilon-1)})\sigma_{\text{avg}}(\pi^o).\end{aligned}$$

Multiplying both sides by τ yields the result. □